

Public Key Crypto notes - 11

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1 Projective coordinates

Recall 1.1. $\mathbb{R}^2$ is the two-dimensional affine plane. In this space, there are:

- Points,
- Lines
- $\mathbb{P}(\mathbb{R}^2)$ is the two-dimensional projective space, that contains:
 - affine points,
 - affine lines,
 - a "projective" point: meeting points of parallel lines,
 - a "projective" line: line of all projective points.

Definition 1.1 (Projective points). For $x_1, y_1, z_1, x_2, y_2, z_2$, where $((x_i, y_i, z_i) \neq (0, 0, 0))$, we say $(x_1, y_1, z_1) \sim (x_2, y_2, z_2)$ if $x_2 = \lambda x_1, y_2 = \lambda y_1, z_2 = \lambda z_1$ for some $\lambda \neq 0$.

This $\sim$ is an equivalence relation. Let $(x : y : z)$ be a representative of the equivalence classes of (x, y, z) .

Example: $(1 : 1 : 1) = (2 : 2 : 2) = (3 : 3 : 3)$, but $(1 : 1 : 1) \neq (2 : 1 : 0)$.

Remark 1.1. If $z \neq 0$, then $(x : y : z) = (\frac{x}{z} : \frac{y}{z} : 1)$. These will be the affine points.

If $z = 0$, then $(x : y : z) = (x : y : 0)$. These will be the projective points.

Definition 1.2 (Homogenous polynomial). A polynomial $F \in \mathbb{R}[x, y, z]$ of form $F = \sum_{c_{i,j,k}} x^i y^j z^k$ is *homogenous* if $c_{i,0,k} \neq 0$, then $i + j + k = d$ for some d . Here, $\deg F = d$.

Examples: $F_1 = c_1 x + c_2 y + c_3$ is not homogenous, while $F_2 = c_1 x + c_2 y + c_3 z$ is $\deg = 1$ homogenous.

Remark 1.2. If F is homogenous and $(x_1 : y_1 : z_1) = (x_2 : y_2 : z_2)$, then $F(x_1, y_1, z_1) = 0 \iff F(x_2, y_2, z_2) = 0$.

Indeed, if:

$$x_2 = \lambda x_1, \quad y_2 = \lambda y_1, \quad z_2 = \lambda z_1. \quad (1)$$

Then

$$F(x_2, y_2, z_2) = F(\lambda x_1, \lambda y_1, \lambda z_1) \quad (2)$$

$$= \sum_{i+j+k=d} c_{i,j,k} (\lambda x_1)^i (\lambda y_1)^j (\lambda z_1)^k \quad (3)$$

$$= \lambda^d \sum_{i+j+k=d} c_{i,j,k} x_1^i y_1^j z_1^k \quad (4)$$

$$= \lambda^d F(x_1, y_1, z_1). \quad (5)$$

Definition 1.3. A line of the projective space is the zero-set of a $\deg = 1$ homogenic polynomial.

Example 1: Take a horizontal line in $\mathbb{R}^2$. $y + c = 0 \rightarrow$ we can homogenize this: $y + cz = 0$.

$(x : y : z)$ is a zero.

- If $z = 1 \rightarrow$ affine point.
- If $z = 0 \rightarrow y = 0 \rightarrow x \neq 0$.

So $(x : 0 : 0) = (1 : 0 : 0)$.

Moreover, $(1 : 0 : 0) = (-1 : 0 : 0)$.

Example 2: Take a vertical line $x + c = 0 \rightarrow$ make it homogenous $\rightarrow x + cz = 0 \rightarrow x + 0y + cz = 0$.

- If $z = 1 \rightarrow$ affine points.
- If $z = 0 \rightarrow x = 0 \rightarrow y \neq 0 \rightarrow (0 : y : 0) = (0 : 1 : 0)$.

1.1 Homogenizing

Let $E : y^2 = x^3 + Ax + B$ and $F = x^3 + Ax + B - y^2$. Let's homogenize F :

$$H = x^3 + Axz^2 + Bz^3 - y^2z$$

The point of E:

- If $z = 1 \rightarrow H(x, y, 1) = 0 \iff (x, y) \in E$.
- If $z = 0 \rightarrow H(x, y, 0) = x^3 \rightarrow x = 0 \rightarrow y \neq 0$.

Thus, $(x : y : 0) = (0 : 1 : 0)$ is the only fully projective point of E .

Remark 1.3.

- A similar addition law can be given, but we do not have to distinguish $P = \infty, P \neq \infty$.
- In implementations, only projective coordinates are used.
- It speeds up computations as modular inversions can be eliminated.

2 Elliptic curve cryptography

Remark 2.1. Let P be a prime, and let E be an elliptic curve over $\mathbb{Z}_p$. Then, the computation $n \rightarrow nP$ can be easily computed using the double-and-add algorithm (where n is an integer, and $P \in E$).

i.e. $n = \sum_{i=0}^{\lambda} b_i z^i, b_i \in \{0, 1\}$.

$nP = 2(2(\dots(2b_i P) \dots) \oplus b_2 P) \oplus b_1 P \oplus b_0 P$.

There are $\approx 2l = 2 \log_2 n$ many curve operations.

But computing n from P, nP is hard. This is called the DLog problem on the curve.¹

Techniques for attack:

1. Brute-force: trying all possible n values. The possible values of $n \approx |E| \approx P$ (if P is not trivial).
2. There are smarter algorithms (like baby-step-giant-step) having a running time of approximately $\sqrt{P}$.
3. Index calculus does *not* work.

Hence, to achieve the same security level AES-128, one needs P to be approximately 2^{256} bit size. For RSA and ElGamal encryption, one needs 3000 bits for the same security level.

2.1 Elliptic Curve Diffie-Hellman key exchange

Let P be a prime and E be an elliptic curve over $\mathbb{Z}_P, P \in E$ of order l (i.e. $lP = \infty$, for $0 < k < l, kP \neq \infty$).

As $E \cong \mathbb{Z}_N$ or $\mathbb{Z}_N \times \mathbb{Z}_M$, E and P can be chosen such that $l \approx P$.

1. Alice chooses $a \in_R \mathbb{Z}_l$ and sends it to Bob.
2. Bob chooses $b \in_R \mathbb{Z}_l$ and sends it to Alice.
3. $a(bP) = abP = b(aP)$.
4. The shared key is $x = abP$

2.2 Elliptic Curve ElGamal encryption

• Gen:

- Let P be a prime and E be an elliptic curve over $\mathbb{Z}_P, P \in E$ of order l .
- $a \in_R \mathbb{Z}_l, Q = aP$.
- $sk = a, pk = (p, E, P, Q)$.

• Enc:

- Given $pk = (p, E, P, Q)$ and let $M \in E$ be a message.
- $c = (rP, M \oplus rQ)$ with $r \in_R \mathbb{Z}_l$.

• Dec:

- Given $sk = a$, and the ciphertext $c = (R, S)$, the message is $M = S \ominus aR$.

¹Where previously, we had g^n as the basis of security, we now have nP for the same purpose.